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Return-to-Play and Performance Outcomes of Professional Athletes in North America After Hip Arthroscopy From 1999 to 2016

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Background: The effect of hip arthroscopy on athletic performance compared with preinjury levels for professional athletes in different sports remains unknown. In addition, while return rates have been reported for professional baseball, football, and hockey players, return rates have not been reported for professional basketball players.

Hypothesis: Professional athletes in 4 major North American sports would be able to return to their sport and preoperative level of performance at a high rate after arthroscopic hip surgery.

Study Design: Descriptive epidemiology study.

Methods: Major League Baseball (MLB), National Basketball Association (NBA), National Football League (NFL), and National Hockey League (NHL) athletes who underwent hip arthroscopy were identified through a previously reported protocol based on public sources. Successful return to play (RTP) was defined as returning for at least 1 professional regular season game after surgery. Performance scores were calculated by use of previously established scoring systems. Each player served as his own control, with the season prior to surgery defined as baseline. To make comparisons across sports, the authors adjusted for expected season and career length differences between sports and calculated percentage changes in performance.

Results: The authors identified 227 procedures performed on 180 professional athletes between 1999 and 2016. Successful RTP was achieved in 84.6% (192/227) of the procedures. Compared with all other athletes, NBA athletes returned at a similar rate (85.7%, $P \geq .999$). NFL offensive linemen returned at a significantly lower rate than all other athletes (61.1%, $P = .010$). NHL athletes returned at a significantly higher rate than all other athletes (91.8%, $P = .048$) and demonstrated significantly decreased performance during postoperative season 1 compared with baseline (–35.1%, $P = .002$). Lead leg surgery for MLB athletes (batting stance for hitters, pitching stance for pitchers) resulted in a 12.7% reduction in hitter performance score ($P = .041$), a 1.3% reduction in pitcher fastball velocity ($P = .004$), and a 60.7% reduction in pitch count ($P = .007$) one season after surgery compared with baseline. Players in nearly every sport demonstrated significant reductions in game participation after surgery.

Conclusion: This study supports the hypothesis that hip arthroscopy in professional athletes is associated with excellent rates of return at the professional level. However, postoperative performance outcomes varied based on sport and position.

Keywords: hip arthroscopy; professional athletes; return to play; performance outcomes

The tremendous physical demands associated with elite levels of play place professional athletes at an increased risk of sustaining a variety of orthopaedic injuries.[9,18,21,28] While injuries involving the hip are less common than other sports injuries—comprising approximately 1% to 6% of all injuries sustained by American professional baseball, basketball, football, and ice hockey players[9,21,22,28]—these injuries nevertheless have the potential to substantially affect the subsequent performance and career longevity of high-level players. As such, it is important that players, team physicians, athletic trainers, and coaches understand the potential career risks and benefits when operative management is indicated for more severe injuries of the hip.

Intra-articular injuries of the hip, including chondrolabral injuries, loose bodies, articular cartilage injuries,

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and femoroacetabular impingement (FAI), are known to cause substantial pain and functional disability in athletes and may force players to retire prematurely.[5,10,22,26] Specific risk factors for developing hip injury or disease, particularly FAI, that have been identified in athletes include repetitive cutting, twisting, and pivoting movements as well as movements requiring a high degree of hip range of motion (eg, as seen in ballet dancers and ice hockey goalies).[5,22] When operative management is indicated, hip conditions are being treated arthroscopically with increasing frequency,[7,16] due to the minimally invasive nature, fewer complications, equal efficacy, and faster recovery of arthroscopic versus open and mini-open techniques.[16,22,26] Despite the frequent use of hip arthroscopy and excellent patient-reported outcomes in the general population, the clinical significance of such improvements after hip arthroscopy remains debated.[16] To this end, the physical stresses to which elite athletes are exposed far exceed those of recreational competitors, warranting separate analyses and potential recommendations.

Recent studies have reported that elite athletes in sports such as ice hockey and football returned from hip arthroscopy at rates of up to 96% and 87%, respectively.[1,22,26] Further, a mixed study of collegiate and professional baseball players found that these athletes returned at a rate of 90%.[4] However, return rates for professional basketball players and the effect that hip arthroscopy has on athletic performance compared with preinjury levels for professional athletes have not been characterized. In addition, no consensus is available on whether certain sports predispose athletes to a higher risk of undergoing hip arthroscopy or worse postoperative outcomes.

The purpose of this study was to evaluate return-to-play (RTP) rates and performance-based outcomes following hip arthroscopy for players of 4 major North American professional sports. We hypothesized that professional athletes returned to their preoperative level of performance at a high rate after surgery. Further, we sought to identify factors associated with successful RTP, postoperative career length, and performance-based outcomes.

METHODS

Inclusion Criteria

Major League Baseball (MLB), National Basketball Association (NBA), National Football League (NFL), and National Hockey League (NHL) athletes who underwent arthroscopic hip surgery were identified through a previously reported algorithmic protocol[12-14,24,29,31] based on public sources. A database of players from each sport who underwent hip arthroscopy was compiled through use of LexisNexis Academic search engine with the following search terms: "[Professional league] arthroscopic hip surgery", "[Professional league] hip labral tear", and "[Professional league] hip impingement". These keywords were chosen because they are commonly used in media reports and would likely be sensitive enough to capture a high number of players who underwent hip arthroscopy. Information collected during this first screening included player name, date of surgery, primary diagnosis (eg, labral tear), and arthroscopic intervention (eg, microfracture). Players and cases captured by the first screening were then validated individually through use of player profiles, newspaper archives, press releases, and team injury reports with more specific search terms: "[Player name] arthroscopic hip surgery", "[Player name] [diagnosis]", and "[Player name] [arthroscopic procedure]". Players identified incidentally at any point during the initial screening or validation process were added to the database and validated using the aforementioned protocol. Inclusion criteria for this study were MLB, NBA, NFL, and NHL athletes who reportedly underwent primary or revision hip arthroscopy to treat symptomatic labral abnormality, chondral defects, loose bodies, and/or FAI while on the active roster of a professional team and for whom a definitive date of surgery was available. Further, for a surgery to be included in this study, at least 1 of the following interventions must have been explicitly cited in reports: labral repair, osteoplasty, debridement, microfracture, and loose body excision. In the event that a player reportedly underwent multiple concomitant interventions on the same hip during the same surgery (eg, a player underwent labral repair and microfracture during the same surgery), the surgery was still counted as a single procedure for that player. Cases were excluded from the database if reports cited hip dislocation, avascular necrosis, sports hernia, muscle injury (eg, hip flexor), hip resurfacing or replacement, or nerve decompression. For each case identified by the initial screen, at least 2 independent public sources of record were cross-referenced to confirm that an arthroscopic hip surgery was performed as defined by the inclusion criteria of this study. Newly drafted players who underwent hip arthroscopy prior to playing in a professional regular season game or rookie players who underwent season-ending or non–season-ending surgery were included only for RTP analysis. Surgeries performed while a player was in college or on a minor league team were excluded. In addition, players who retired before surgery, players for whom postoperative data were not available (eg, those who underwent surgery too recently), or cases that had conflicting reports from different sources were also excluded.

Performance-Based Outcome Measures

Demographic data were manually collected for each athlete who met the inclusion criteria, including date of birth, weight, height, date of professional debut, position, date of most recent professional regular season game, and retirement date (when applicable). Successful RTP was defined as returning for at least 1 professional regular season game after surgery. This definition was chosen primarily because differences in the physical demands and skill level required for professional-level play compared with minor or international league play could introduce confounders if these other levels of play were included. Additionally, information regarding professional game participation is more readily and consistently available from public sources compared with other possible return points (eg, first

TABLE 1
Performance Score Calculations for National Football League (NFL) Players Based on Position[a]

	Performance Score
Quarterback	[(Passing Yards/25) + (Rushing Yards/10) + (4 × Touchdowns) – (2 × Interceptions)]/Games Played
Running back	[(Total Yards Gained/10) + (6 × Touchdowns)]/Games Played
Wide receiver/tight end	[(Total Yards Gained/10) + (6 × Touchdowns)]/Games Played
Defensive player	[(Sacks) + (2 × Interceptions) + (6 × Touchdowns)]/Games Played
Kicker	[(3 × Field Goals Made) + Extra Points Made]/Games Played

[a]Statistics to calculate performance score were not available for offensive linemen (including long snappers) and punters, and these players were excluded from performance evaluation. Data from Hsu.[12]

TABLE 2
Performance Score Calculations for National Hockey League (NHL) Players Based on Position[a]

	Performance Score
Center/wing	[(3 × Even-Strength Goals) + (2 × Power Play Goals) + (4 × Short-Handed Goals) + (4 × Game-Winning Goals) + (2 × Assists) + (plus/minus) – (0.25 × Penalty Minutes) – (0.33 × Shots on Goal)]/Games Played
Defenseman	[(5 × Even-Strength Goals) + (4 × Power Play Goals) + (6 × Short-Handed Goals) + (5 × Game-Winning Goals) + (3 × Assists) + (plus/minus) – (0.25 × Penalty Minutes) – (0.33 × Shots on Goal)]/Games Played
Goalie	[(0.7 × Wins) + (0.2 × Ties Plus Overtime Losses) + Shutouts + (0.17 × Saves) – (0.25 × Losses) – (1.23 × Goals Against)]/Games Played

[a]Data from Schroeder et al.[32]

postoperative workout, practice, or scrimmage). Return time was defined as the number of days elapsed between reported date of surgery and date of first professional game after surgery. For season-ending and non–season-ending surgeries, the index season was defined as the professional regular season before the season in which surgery occurred. For off-season surgeries, the index season was defined as the professional regular season immediately before surgery. For season-ending and off-season surgeries, the season corresponding to a player's first professional game back was considered postoperative season 1. For non–season-ending surgeries, the professional season after the season in which surgery occurred was considered postoperative season 1. Both pre- and postoperative regular season player statistics were compiled for each player, including all-star appearances, years and games played, games started (except hockey players), and performance score (PS). Two statistical time points were used after hip arthroscopy: postoperative season 1 and an average of postoperative seasons 2 and 3. Each player served as his own control, with the season prior to surgery (index season) used as baseline.

Sport-specific performance scores were calculated for players who successfully returned based on previously established scoring systems that have been refined over the past 8 years. For baseball, performance scores were compiled by use of accepted sabermetrics based on a player's role as a hitter or pitcher. The number of walks plus hits per inning pitched was used as the PS for pitchers,[11] and on-base plus slugging percentage was used as the PS for hitters.[15] For basketball, player efficiency rating was used as the primary performance outcome.[3,6] Sport-specific scores were manually computed for football[12] and hockey[32] players (Table 1 and Table 2, respectively).

Statistics to calculate PS were not available for rookies or first-year players and non–skill-position football players (offensive linemen, long snappers, and punters), and these players were excluded from the performance evaluation. The ratio of games started to games played (GS:GP) was calculated for each player as an additional measure of player performance.[13] Further, for MLB pitchers, 4-seam fastball velocity and pitch counts were manually collected for players' index season and up to 3 seasons postoperatively from the PITCHf/x public dataset (Sportvision, Inc), which has digitally tracked and recorded every MLB pitch since 2007.[2] Subgroup analysis was performed on MLB pitchers and hitters who underwent surgery on their lead leg (pitching stance for pitchers, batting stance for hitters) as well as NHL goalies in order to evaluate whether these positions portend worse outcomes after hip arthroscopy, as they involve characteristically provocative movements associated with intra-articular hip injury or disease (eg, extreme flexion, adduction, and internal rotation of the hip).[4,5,10,23,27,30,33]

For players who met inclusion criteria for PS evaluation, a dose-response analysis of return time and change in PS was performed to assess what an appropriate range of time for return from hip arthroscopy may be. Three return time cutoffs were imposed to obtain greater resolution of the dose-response relationship: (1) the mean return time, (2) 1 SD below the mean return time, and (3) 120 days. The third cutoff was chosen based on a previous case series that reported professional hockey players were cleared to participate in their first game an average of 3.9 months after hip arthroscopy.[26] Players were divided into 2 groups based on whether they returned before or after the cutoff. Analysis was performed on the combined cohort as well as on individual sports.

TABLE 3
Number of Hip Procedures Performed and Return to Play (RTP) Based on Player Position for Professional Athletes Identified in this Study[a]

Sport	Position	Return/Total Hips, No.	% RTP
Major League Baseball	Pitcher	22/26	84.6
	Hitter	26/33	78.8
National Basketball Association		24/28	85.7
National Football League	Defense	19/22	86.4
	Quarterback	3/4	75.0
	Running back	3/3	100.0
	Wide receiver/tight end	12/13	92.3
	Kicker	3/4	75.0
	Offensive lineman[b]	11/18	**61.1**[c]
	Long snapper[b]	1/2	50.0
	Punter[b]	1/1	100.0
National Hockey League	Forward (center/winger)	36/38	94.7
	Defenseman	13/14	92.9
	Goalie	18/21	85.7

[a]Values reflect the number of hips operated on rather than the number of individual athletes. Positions listed correspond to the positions used in calculating performance scores.
[b]Excluded from performance score evaluation.
[c]Bold value denotes significance compared with players in other sports identified in this study who underwent hip arthroscopy ($P = .010$).

Comparisons across sports were made by adjusting for expected season and career length differences between sports and by calculating percentage changes in performance. Number of games played was normalized for each player to account for differences in typical regular season length between sports (MLB, 162 games; NBA, 82 games; NFL, 16 games; NHL, 82 games). Career lengths were standardized for each player by adjusting for differences in expected career length between sports (MLB, 5.6 years; NBA, 4.8 years; NFL, 3.5 years; NHL, 5.5 years) based on a previously published method.[14] For each player in each sport, games played and career length were normalized as follows: {[Total Number of Regular Season Games Played by (or Career Length of) Athlete] / [Total Regular Season Length (or Average Career Length) in Athlete's League]} × Reference League Regular Season Length (or Average Career Length). MLB was arbitrarily chosen as the reference for regular season length and average career length. This study was deemed exempt by the institutional review board as all data were obtained from publicly available sources.

Statistical Analysis

IBM SPSS statistics software (version 24.0; IBM Inc) was used for all statistical analyses. Continuous variables were evaluated through use of a 2-tailed Student t test for normally distributed data. A Fisher exact test was performed to evaluate categorical variables. Each player served as his own control, and changes in pre- and postoperative player statistics were assessed by use of a 2-tailed paired t test. Regression models were used to evaluate the effect of independent variables (age, experience, and body mass index [BMI]) on RTP and performance outcomes. Kaplan-Meier survivorship analysis was performed with the endpoint defined as official retirement from sport. For all statistical analyses, the threshold for significance was established at $P \leq .05$. Unless noted otherwise, numerical values are expressed mean ± SD, and P values correspond to each sport or player subgroup compared with all other sports or players combined. Where reported, a comparison with "all other players" indicates comparison with players in other sports identified in this study who underwent hip arthroscopy, rather than players in the same sport.

RESULTS

In total, 227 arthroscopic hip procedures (59 MLB, 28 NBA, 67 NFL, 73 NHL) performed on 180 professional players (47 MLB, 24 NBA, 54 NFL, 55 NHL) between 1999 and 2016 were identified (Table 3). Labral repair was reported in 73.1% (166/227) of the cases identified. Other reported procedures included osteoplasty in 22.9% (52/227) of cases, debridement in 9.3% (21/227) of cases, loose body excision in 7.9% (18/227) of cases, and microfracture in 5.7% (13/227) of cases. Of the 227 total procedures, 30 (13.2%) were performed on players who had bilateral procedures (average time between procedures, 18.4 months), 17 (7.5%) were performed on players who had revision procedures (average time between procedures, 20.2 months), and 180 (79.3%) were single, unilateral procedures (including the first surgery for players who subsequently underwent bilateral and/or revision hip arthroscopy).

Player Demographics at Time of Surgery

The mean age at the time of surgery for all players was 28.9 ± 4.2 years, which varied according to sport. MLB

TABLE 4
Summary of Player Demographics and Preoperative Career Statistics for Professional Athletes Included in this Study, Stratified by Sport Played

	Major League Baseball (n = 59)	National Basketball Association (n = 28)	National Football League (n = 67)	National Hockey League (n = 73)
Age, y	30.0 ± 3.8	26.6 ± 3.8	28.5 ± 3.8	29.4 ± 4.6
Players in other sports, y	28.6 ± 4.2	29.3 ± 4.1	29.1 ± 4.3	28.7 ± 4.0
P value	**.022**[a]	**.001**	.266	.223
Body mass index, kg/m^2	26.5 ± 2.6	24.4 ± 1.2	32.0 ± 4.6	26.3 ± 1.5
Players in other sports, kg/m^2	28.3 ± 4.4	28.3 ± 4.1	26.0 ± 2.1	28.5 ± 4.7
P value	**.004**	**<.001**	**<.001**	**<.001**
Career before surgery, y[b]	6.6 ± 4.6	6.3 ± 3.8	8.5 ± 5.3	8.7 ± 4.9
[unadjusted]	—	[5.4 ± 3.2]	[5.3 ± 3.3]	[8.5 ± 4.8]
Players in other sports, y	8.3 ± 5.0	8.0 ± 5.0	7.5 ± 4.7	7.4 ± 4.9
P value	**.029**	.107	.153	.082
Career before surgery, No. of games[b]	534 ± 644	633 ± 410	793 ± 504	874 ± 630
[unadjusted]	—	[321 ± 203]	[78 ± 49]	[442 ± 317]
Players in other sports, No. of games	804 ± 554	744 ± 609	705 ± 624	665 ± 561
P value	**.003**	.388	.314	**.014**
GS:GP[c] before surgery[d]	0.65 ± 0.37	0.45 ± 0.34	0.64 ± 0.34	
Players in other sports	0.59 ± 0.35	0.64 ± 0.35	0.59 ± 0.37	
P value	.290	**.017**	.475	

[a]Bold values denote significance compared with players in other sports identified in this study who underwent hip arthroscopy.
[b]Career length and number of games played before surgery were adjusted to account for differences in average expected career length and regular season length between sports, with Major League Baseball arbitrarily chosen as the reference for average career length and regular season length.
[c]Ratio of games started to games played.
[d]Game starts are not compiled for National Hockey Leage athletes; thus, the ratio of games started to games played (GS:GP) was not calculated for these players.

players were significantly older than all other players (*P* = .022) and NBA players were significantly younger (*P* = .001) (Table 4). MLB players also had significantly shorter preoperative career lengths (years and games) than players in all other sports combined (*P* = .029 and *P* = .003, respectively). Conversely, NHL athletes played significantly more games before surgery than all other players (*P* = .014). NBA athletes demonstrated a lower average total GS:GP before surgery than all other players (*P* = .017).

Return to Play

Overall, successful RTP was achieved in 84.6% (192/227) of the procedures. A significantly higher proportion of NHL players returned after surgery than players in all other sports combined (91.8% vs 81.2%, *P* = .048) (Figure 1). Subgroup analysis revealed RTP rates of 83.3% (10/12) for MLB pitchers who underwent surgery on their lead leg, 70.6% (12/17) for MLB hitters who underwent surgery on their lead leg, and 85.7% (18/21) for NHL goalies, none of which were significant compared with all other players. In addition, NFL offensive linemen returned from 61.1% (11/18) of procedures, which was significantly lower than the rates for all other players (*P* = .010). No statistically significant difference was found in RTP rate after hip arthroscopy with and without microfracture (9/13 [69.2%] vs 183/214 [85.5%], respectively; *P* = .121). Players with a history of bilateral hip arthroscopies returned at a rate similar to players who underwent a single, unilateral procedure (26/30 [86.7%] vs 156/180 [86.7%]; *P* ≥ .999). Players who underwent revision hip arthroscopy returned at a significantly lower rate than players who underwent a single, unilateral procedure (10/17 [58.8%] vs 156/180 [86.7%]; *P* = .008). When cases were stratified based on

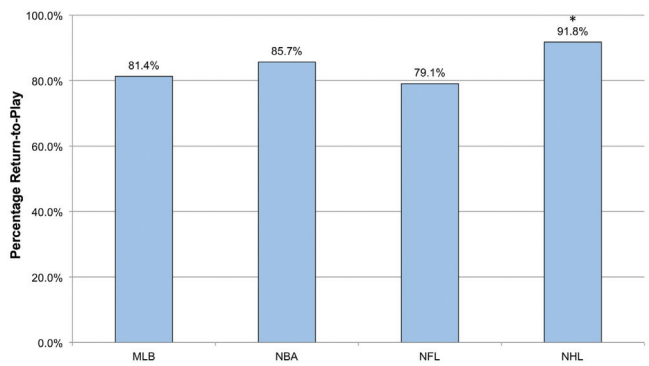

Figure 1. Return-to-play rates following hip arthroscopy in athletes of 4 major North American professional sports. *NHL athletes returned at a significantly higher rate than the average of all other athletes (91.8% vs 81.2%, *P* = .048). MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

TABLE 5
Comparison of Player Age and Career Statistics Between Players Who Returned to the Professional Level After Hip Arthroscopy and Players in the Same Sport Who Did Not, Stratified by Sport[a]

	RTP	No RTP	*P* Value[b]
Age, y			
Overall	28.5 ± 4.0	31.3 ± 4.3	**<.001**
MLB	29.5 ± 3.4	32.3 ± 4.6	**.022**
NBA	26.0 ± 3.7	30.0 ± 2.3	**.05**
NFL	27.7 ± 3.4	31.4 ± 3.8	**.001**
NHL	29.4 ± 4.5	29.9 ± 6.2	.785
Career before surgery, y			
Overall[c]	7.4 ± 4.6	10.0 ± 5.9	**.004**
MLB	6.0 ± 4.0	9.3 ± 6.1	**.034**
NBA	4.8 ± 3.1	8.5 ± 2.5	**.032**
NFL	4.7 ± 3.0	7.6 ± 3.6	**.003**
NHL	8.7 ± 4.7	6.4 ± 6.8	.264
Index season games, n			
Overall[c]	110 ± 51	73 ± 58	**<.001**
MLB	76 ± 48	39 ± 46	**.032**
NBA	48 ± 28	31 ± 3	.229
NFL	15 ± 3	12 ± 4	**.013**
NHL	59 ± 22	10 ± 11	**<.001**

[a]Unless noted otherwise, tabulated preoperative career lengths and index season games played reflect raw (unadjusted) values. MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League; RTP, return to play.

[b]Bold values denote significance between players who returned and those in the same sport who did not.

[c]Reflects adjusted values to account for differences in average expected career length and regular season length between sports, with MLB arbitrarily chosen as the reference for average career length and regular season length.

whether the surgeries were performed during the first half of the study period (1999-2007) versus the second half (2008-2016), fewer cases were reported during the first half (n = 48) compared with the second half (n = 179), but overall return rates were comparable (85.4% vs 84.4%, respectively; $P \geq .999$).

Overall, players who returned were significantly younger at the time of surgery ($P < .001$), played fewer preoperative years ($P = .004$), and played more games during their index season ($P < .001$) compared with players who did not return (Table 5). Controlling for individual differences in player age and preoperative career experience (years), binary logistic regression analysis confirmed that for every game in which an athlete participated during his index season, he had an approximately 2% higher probability of returning after hip arthroscopy (odds ratio, 1.02; 95% CI, 1.01-1.02; $P < .001$). NHL players who returned demonstrated significantly longer total unadjusted career lengths compared with NHL players who did not return (11.4 ± 4.7 years vs 6.2 ± 6.7 years, respectively; $P = .014$). Overall, differences in BMI, GS:GP (index season and total preoperative career), and games played before surgery between players who did and did not return within the same sport were not significant.

Average RTP time was 209 ± 132 days, with NHL players returning significantly sooner than players in all other sports combined ($P = .014$) (Table 6). Multiple linear regression analysis found that more games played during the index season was predictive of a shorter return time ($\beta = -0.266$, $R = -0.266$, $P < .001$). No statistically significant difference was found in RTP time after hip arthroscopy with and without microfracture (284 ± 246 days vs 205 ± 125 days, $P = .082$). MLB players who returned played significantly fewer games after surgery compared with all other players ($P = .004$). Conversely, NFL players who returned had significantly longer postoperative careers (years and total games) than all other players ($P = .019$ and $P = .006$, respectively). With respect to postoperative career length (years and total games), multiple linear regression analysis found player age to be a significant negative independent predictor ($\beta = -0.233$, $R = -0.218$, $P = .001$, and $\beta = -0.161$, $R = -0.192$, $P = .004$, respectively) and games played during index season to be a positive independent predictor ($\beta = 0.192$, $R = 0.173$, $P = .007$, and $\beta = 0.368$, $R = 0.383$, $P < .001$, respectively).

Kaplan-Meier survivorship analysis demonstrated that 81% and 62% of players who returned were expected to still be on active rosters in their respective leagues at the 1- and 3-year time points after surgery, respectively, with an overall median (50%) survival of 4.2 years after hip arthroscopy and no significant differences across sports (Figure 2). NBA players had the highest median survival (7.4 years), while MLB and NHL players had the lowest median survival (3.5 years for both).

Performance Evaluation

Of the 192 cases identified by this study in which successful RTP was achieved, 170 (88.5%) met the inclusion criteria for PS evaluation (48 MLB, 20 NBA, 40 NFL, 62 NHL) and had at least 1 year of follow-up data available, with 119 (62.0%; 32 MLB, 15 NBA, 32 NFL, 40 NHL) having 2 or more years of follow-up data available. In total, 7 rookie players (4 NBA, 3 NHL) and 13 NFL players (11 offensive linemen, 1 punter, 1 long snapper) who returned did not have statistics available for calculation of PS and were excluded from performance evaluation. Additionally, 2 NHL players underwent non–season-ending surgery and retired after returning (before postoperative season 1 as defined by this study). Overall, players in this study demonstrated a significant decrease in PS in their first postoperative season (–15.8%, $P = .022$) and at postoperative seasons 2 and 3 (–14.4%, $P = .008$) relative to their index season. Analysis of individual sports revealed that changes in PS were not significant for MLB (overall), NBA, or NFL players (Figure 3). Only NHL players demonstrated a significant decrease in PS compared with their index season, which occurred during postoperative season 1 (–35.1%, $P = .002$) and was recoverable to baseline by postoperative seasons 2 and 3. Subgroup analysis revealed that MLB hitters who underwent surgery on their lead leg demonstrated a significantly reduced PS 1 season after surgery compared

TABLE 6
Summary of Return to Play (RTP) Times and Postoperative Career Lengths for Professional Athletes Who Successfully Returned to Play, Stratified by Sport Played

League	RTP Time, d	Career After Surgery, y [Unadjusted][a]	Career After Surgery, No. of Games [Unadjusted][a]
Major League Baseball	222 ± 118	2.8 ± 2.6	192 ± 257[b]
National Basketball Association	243 ± 196	3.7 ± 3.6 [3.1 ± 3.0]	371 ± 369 [188 ± 183]
National Football League	222 ± 116	4.1 ± 3.5[c] [2.5 ± 2.2]	413 ± 354[d] [41 ± 35]
National Hockey League	177 ± 121[e]	2.7 ± 2.6 [2.7 ± 2.5]	288 ± 301 [144 ± 154]

[a]Career length and number of games played after surgery were adjusted to account for differences in average expected career length and regular season length between sports, with Major League Baseball arbitrarily chosen as the reference for average career length and regular season length.
[b]Denotes significance compared with all other players (348 ± 336 games, P = .004).
[c]Denotes significance compared with all other players (2.9 ± 2.8 years, P = .019).
[d]Denotes significance compared with all other players (269 ± 304 games, P = .006).
[e]Denotes significance compared with all other players (226 ± 136 days, P = .014).

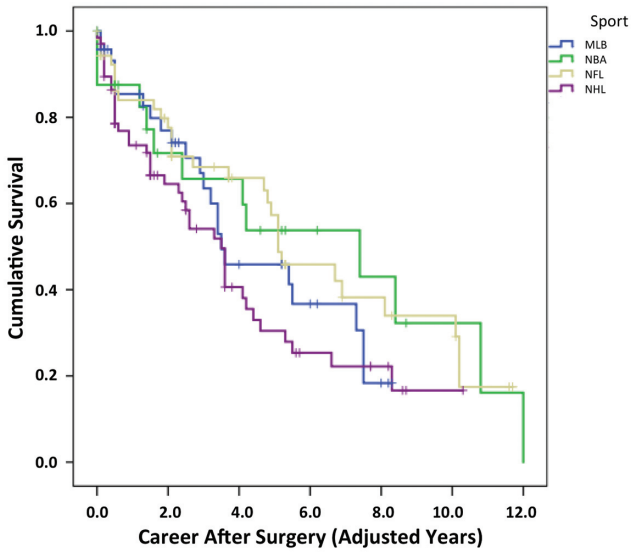

Figure 2. Kaplan-Meier survivorship analysis for professional athletes who underwent hip arthroscopy, with survivorship defined as official retirement from sport. Career length after surgery was standardized for each player by adjusting for differences in expected career length between sports. Highest median survival occurred among NBA players (7.4 years) and lowest median survival occurred among MLB and NHL players (3.5 years for both). MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

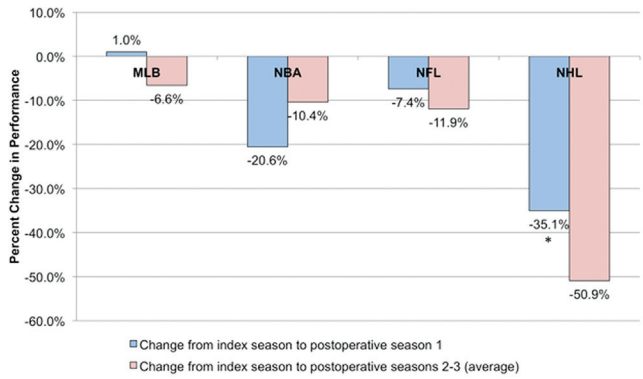

Figure 3. Performance-based outcomes following hip arthroscopy in professional athletes in the short term (postoperative season 1) and medium term (average of postoperative seasons 2 and 3). *Denotes significance compared with index season (P = .002). MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

with index (–12.7%, P = .041), which continued into postoperative seasons 2 and 3 (–9.9%, P = .025); no significant changes in PS were found for MLB pitchers who underwent surgery on their lead leg or NHL goalies. Dose-response analysis identified no significant associations between RTP time and change in PS.

Additional analysis of players who met inclusion criteria for PS evaluation revealed that relative to their index season, players in nearly every sport played significantly fewer games, had significantly fewer starts, and had a significantly lower GS:GP after surgery (Figure 4). Further, MLB pitchers had significantly decreased average 4-seam fastball velocity during postoperative season 1 compared with index (–1.3%, P < .001), which was recoverable to baseline by postoperative seasons 2 and 3. Analysis of MLB pitchers who underwent surgery specifically on their lead leg demonstrated a similar reduction in average 4-seam fastball velocity (–1.3%, P = .004) and a significantly decreased pitch count during postoperative season 1 compared with index (–60.7%, P = .007); both findings were recoverable to baseline by postoperative seasons 2 and 3.

DISCUSSION

The data in this study demonstrate excellent RTP outcomes (84.6%) for professional athletes after hip arthroscopy. Significant differences were found in RTP rates based on sport,

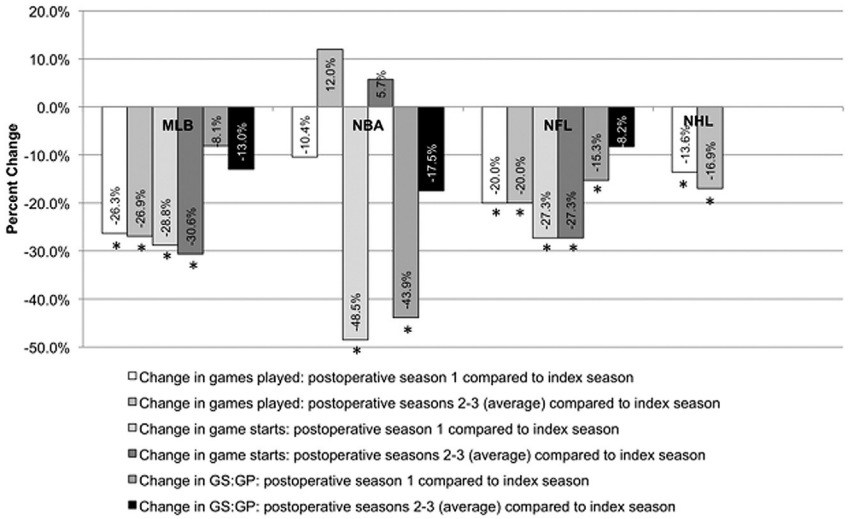

Figure 4. Comparison of preoperative and postoperative career statistics for athletes of 4 major North American professional sports who returned to play after hip arthroscopy. *Denotes significance compared with index season, $P \leq .05$. GS:GP, ratio of games started to games played; MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

with NHL players returning at a significantly higher rate (91.8%) and significantly sooner than all other players. Our findings are in concordance with previous studies, which have reported return rates from hip arthroscopy ranging from 77% to 86% for mixed groups of elite athletes,[1,20] 50% to 90% for mixed groups of elite baseball players,[4,25] 80% to 87% for NFL players,[22,25] and 82% to 100%[19,23,26] for NHL players, depending on the specific procedures performed. Our results are also partially consistent with previous studies which reported that players in contact sports (eg, football, ice hockey) did not have an increased risk of not returning to their sport after hip arthroscopy,[19,22] although we found that NFL offensive linemen specifically may be at an increased risk of not returning.

We found no statistically significant difference in RTP rates between players who underwent hip arthroscopy with and without microfracture, which is consistent with the findings of previous retrospective studies involving elite athletes.[17,20] In addition, although the surgical trauma involved in microfracture might be expected to require a longer recovery time, we found no evidence that microfracture delayed RTP compared with cases where microfracture was not performed. Limited data are available comparing RTP times between sports following microfracture for hip chondral defects, and although further research is warranted, our findings lend further support to the viability of the microfracture technique for the treatment of full-thickness chondral defects of the hip in elite athletes.

Previous studies have reported that the hips of ice hockey goalies are subjected to a variety of demanding sport-specific postures, especially with regard to the popular butterfly-style save, which may predispose these players to a higher risk of symptomatic FAI and overuse injuries.[10,23,27,30,33] Similarly, the leading hip of baseball pitchers and hitters works to generate tremendous power during wind-up or batting and may predispose these athletes to a higher risk for intra-articular hip injuries and the need for operative management.[4] Despite these purported risks, we observed no significant differences in return rates for MLB pitchers or hitters who had surgery on their lead leg or for NHL hockey goalies compared with all other players. These findings suggest that despite having a potentially higher risk for developing hip injuries, these players have an equal chance of returning after hip arthroscopy relative to all other players. Interestingly, we found that NFL offensive linemen returned at a significantly lower rate (61.1%) compared with all other players, which is consistent with a previous case series and may be secondary to the increased BMI (eg, more advanced chondral degeneration) and repetitive push-off from the line with the hips flexed in these players.[22]

Players in the current study who returned to sport were younger at the time of surgery, played fewer years professionally before surgery, and played in more games during index season compared with players who did not return. A higher number of games played during the index season was also associated with a shorter return time. While the trends observed with respect to years and games played before surgery may seem counterintuitive, the number of years played reflects how long an athlete played professionally whereas the number of games played reflects the athlete's actual level of participation and thus can be viewed as a surrogate for an athlete's role on his team.[18] It is reasonable to speculate that an athlete who participated in more games preoperatively likely has a more prominent role on the team and may have more incentive to return than an athlete who played in fewer games preoperatively.[18]

Performance-based outcomes following hip arthroscopy varied significantly depending on the sport. Hockey players demonstrated a significant reduction in PS one season after surgery compared with index, which was recoverable to baseline within 3 seasons postoperatively. Conversely, baseball (pitchers and hitters combined), basketball, and football players in this study did not demonstrate any significant changes in PS after surgery compared with baseline. Of note, NHL goalies did not demonstrate a significant decrease in PS after surgery compared with baseline. Finally, this study identified that player participation (games played, game starts, and GS:GP) may be expected to decrease postoperatively compared with index, but whether (and when) participation levels return to baseline may depend the sport.

No consensus is available regarding when it is appropriate for players to return to elite-level sports following hip arthroscopy. A previous single-surgeon case series concluded that players may be allowed to return based on their level of pain and hip function.[26] The referenced study reported that professional hockey players were cleared for game participation an average of 3.9 months after surgery, approximately 2 months earlier than hockey players in the present study participated in their first professional game after surgery (5.8 months after surgery, on average). A likely explanation for why apparent return times differed between the present study and the previous case series is that the referenced study evaluated medical clearance whereas the present study evaluated game play. Although a player may be medically cleared to play based on his level of pain and function, he may not yet be recovered to the extent necessary to perform at a high level. Additionally, if a player were medically cleared for game participation during the offseason, he would have to wait until the season began to actually play. We found that hockey players in the present study returned to professional game play, on average, approximately 7 weeks sooner than players in all other sports and demonstrated significantly reduced performance scores during the season after surgery. By returning more quickly, hockey players may have had to adopt less aggressive playing styles and/or experienced hip discomfort that could have reduced performance in the short term (one season after surgery) but did not affect longer-term performance outcomes. Additionally, it may be that the daily collisions, hard playing surfaces, and repetitive hip motions of hockey players could lead to a greater risk of symptoms from a surgically repaired hip and may have a more profound impact on athletic performance compared with noncontact sports (eg, baseball) or sports with fewer provocative hip motions (eg, basketball, football). Notwithstanding, we did not find any significant association between return time and change in PS after surgery. Hence, our data suggest that RTP time is not the sole determinant of postoperative performance and that players who return more quickly are not necessarily at risk of worse performance. Instead, the optimal return time following hip arthroscopy is likely highly individualized and dependent on a number of factors, including extent of hip injury, time from injury to surgery,[26] type of arthroscopic procedures performed, sport, position, and the ability of a player to perform and/or modify sport-specific tasks and maneuvers as necessary upon returning. Correspondingly, we recommend that these factors be closely evaluated, in addition to assessment of a player's function and level of pain, when one is determining a player's readiness for return to game participation.

For MLB pitchers and hitters, the hips are part of a complex kinematic linkage involved in transferring force from the legs to the torso and ultimately into ball or bat speed.[4] Analysis of MLB hitters who underwent surgery on their lead leg demonstrated a significantly reduced PS of approximately 10% to 13% up to 3 seasons postoperatively compared with index. A possible explanation for this finding is that the lead batting leg is forced into internal rotation during the swing, which may result in hip pain that causes poor compensatory hip mechanics and ultimately reduced batting performance for these athletes. No significant changes in PS were found for MLB pitchers, and while the observed reduction in average 4-seam fastball velocity was statistically significant, one could argue that the result (an approximately 1.2 miles per hour decrease in velocity for a pitcher who throws 95 miles per hour on average) was not clinically significant. However, analysis of MLB pitchers who specifically underwent surgery on their lead leg demonstrated a significant reduction in pitch count during the first postoperative season. One possible reason for the observed decrease in pitch velocity may be that a recovering hip is less able to efficiently generate and transfer power.[8] Consequently, a pitcher may need to rely more on his trunk and upper extremity to generate power in the short term after surgery[8]; depending on the player's ability to compensate, he may experience only a slight decrease in velocity. The integral role of the hips in generating power during the pitching motion and the need to compensate with the upper body due to an injured or recovering hip is a proposed reason for why MLB pitchers who undergo hip arthroscopy have a 7-fold higher incidence of undergoing ulnar collateral ligament reconstruction.[4] Therefore, our finding that MLB pitchers recovering from hip arthroscopy threw significantly fewer pitches during their first postoperative season is entirely reasonable, as this would allow pitching mechanics to regain proper form without exposing the upper extremity to undue stress that could potentially result in the need for additional surgery.

Other, more intangible reasons may be responsible for the observed sport-specific differences in RTP and performance outcomes for athletes included in this study. Factors such as individual playing styles, coaching changes, and the realities of contract negotiations can have a considerable impact on a player's role within a team and his incentive to return after surgery. Structural differences in the professional leagues included in this study, such as having a "minor league" for every sport except football, may give certain players a better chance of returning to their previous level. Finally, the variability in playing opportunities based on the number of players on a team and in a game for a given sport may also account for some of these differences.

This study has several limitations that are inherent to a retrospective study and that have been documented in previous publications.[12-14,18,24] The capture rate of players who underwent arthroscopic hip surgery comes into question when public sources are used. For example, individual policies and preferences of leagues, teams, and players may have introduced sampling bias toward more high-profile players. However, with the greater media attention that hip arthroscopy has received and the increased understanding of the layperson regarding procedures such as labral repair and microfracture, we believe that players included in this study are representative of those who have undergone hip arthroscopy. Our method of using publicly available data is the only way to evaluate surgical outcomes in this patient population aside from single-surgeon case series, which themselves may be limited in size and unable to be compared across multiple professional sports. As with other studies that have used public sources, our study is subject to the potential for errors and omissions in the available data, which may have introduced confounders and influenced the results. To this end, we limited the study period to be recent enough that more reliable, redundant reporting was available. Each case was extensively screened to confirm the exact procedures that were reportedly performed, and any surgeries in which significant variability could be assumed without the aid of an operative report were excluded. For example, players who reportedly underwent hip arthroscopy but for whom an exact procedure was not cited were excluded from this study, as the variety of possible interventions could affect recovery time and/or performance outcomes. Similarly, data regarding diagnosis (eg, presenting symptoms, cause, severity, diagnostic imaging), persistence of symptoms, use of conservative therapy, specific surgical indications, intraoperative details, postoperative management and rehabilitation compliance, follow-up examinations (eg, imaging, assessment of readiness/clearance for return), and complications were not available given our method of using public sources. The smaller sample sizes of certain arthroscopic procedures in certain sports were not amenable to statistical analysis, and therefore outcomes within individual sports were not compared based on procedure. More detailed analyses of outcomes by position were not possible, particularly for baseball and basketball, given that players may switch positions in a game, during the season, and throughout their careers. Revision and bilateral cases were included, similar to previous studies,[1,13,26,32] which may have influenced the results. A group of position- and age-matched controls who did not undergo hip arthroscopy was not included in this study, due to limited reporting of specific injuries or diseases for these players. Players included in this study who underwent hip arthroscopy after 2014 may lack follow-up data for postoperative seasons 2 and 3, which could have influenced the results for these time points. Finally, this study evaluated only male professional athletes, which may not translate to the female professional athlete population.

The findings of this study support part of our hypothesis, namely, a high rate of return to professional play after hip arthroscopy. However, the study results also contradict part of our hypothesis in that performance outcomes after hip arthroscopy were less consistent. The ability to provide quantitative, objective information regarding the likely effect of hip arthroscopy on a professional athlete's career is highly desirable for those who provide care. Our study found that NHL players may be able to return quickly from hip arthroscopy at a high rate, but return to preoperative performance and game participation may be an uphill battle for these players. Similarly, while MLB (pitchers and hitters combined), NBA, and NFL players did not demonstrate any significant reductions in PS after surgery, these players generally had lower return rates and slower return times compared with NHL players and participated in fewer games after surgery compared with their index season. These findings are likely attributable at least in part to differences in the relative physical demands associated with each sport and with certain positions. To our knowledge, this is the only study of its kind comparing RTP rates and performance outcomes following hip arthroscopy for professional athletes in American baseball, basketball, football, and ice hockey. The trends identified by this study may help clinicians to manage athlete expectations and guide selection of patients for whom hip arthroscopy has demonstrated more consistent, positive outcomes.

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